

GroundedNutriRec Technical Report: Week 1 Foundation and Dataset Exploration

Framework: GroundedNutriRec: Retrieval-Augmented Multi-Objective LLM Framework for Health-Aware and Explainable Food Recommendation

Report Type: Academic Technical Paper (Week 1 Review)

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1. Introduction

Food recommendation systems are critical tools in modern digital health, assisting users in discovering recipes that align with their personal taste preferences and nutritional needs. Traditional recommendation models focus primarily on predicting user rating preferences based on historical interaction patterns (collaborative filtering) or recipe descriptions (content-based filtering). However, these systems often neglect nutritional quality, leading to recommendations that may reinforce unhealthy eating habits.

To address this limitation, the GroundedNutriRec framework is designed as a Retrieval-Augmented Multi-Objective Large Language Model (LLM) recommendation system. It integrates user preference learning with dietary guidelines (such as WHO nutritional thresholds) and generates natural language explanations grounded in verified recipe metadata. This report summarizes the Week 1 foundations, focusing on literature review, dataset schemas, and comprehensive exploratory data analysis (EDA) of the Food.com corpus.

2. Literature and Architectural Review

To build a health-aware and explainable recommendation system, we review five core sub-systems from a computer science perspective:

2.1 Collaborative Filtering and Recommendation Baselines

Collaborative filtering (CF) relies on the user-item interaction matrix to identify patterns of shared preferences. In food recommendation, implicit feedback (e.g., whether a user "liked" a recipe based on their rating threshold) is typically preferred over explicit ratings due to the high sparsity of the interaction matrix. Baselines include popularity-based filters, content-based recommendation via TF-IDF or text embeddings, and Matrix Factorization (e.g., Singular Value Decomposition).

2.2 Multi-Objective Optimization in Ranking

Ordinary recommender systems maximize a single accuracy metric (e.g., Click-Through Rate or Precision). GroundedNutriRec utilizes multi-objective optimization to balance:

1. User Taste Preference (predicted rating or similarity score)
2. Nutritional Healthiness (WHO dietary density score)
3. Cooking Time / Complexity (as a proxy for user convenience)
4. Recipe Popularity and Novelty

2.3 Retrieval-Augmented Generation (RAG) for Explainability

While LLMs can generate persuasive text, they are prone to hallucinations (e.g., fabricating cooking times or claiming a high-sodium recipe is healthy). GroundedNutriRec implements a RAG pipeline that indexes recipe details into a vector database (using FAISS and SentenceTransformer embeddings). The LLM is forced to generate explanations using only the retrieved evidence, ensuring explanations are factually accurate.

2.4 Natural Language Inference (NLI) for Faithfulness Verification

To guarantee the reliability of the recommendations, explanations are decomposed into individual claims. A Natural Language Inference (NLI) model assesses whether each claim is logically entailed by the retrieved recipe metadata (the evidence). The faithfulness score is calculated as:

$$\text{Faithfulness} = \text{Supported Claims} / \text{Total Claims}$$

3. Dataset Characterization and Schemas

Two primary datasets are explored for the GroundedNutriRec framework: the Food.com dataset (Recipes and Reviews) and the RecipeNLG dataset.

3.1 Food.com Dataset

The Food.com dataset is the primary source for interaction history and recipe metadata. It consists of two main tables:

1. **RAW_recipes.csv:** Metadata for 231,637 recipes, including ingredients, preparation steps, preparation time, and a 7-element nutrition vector.
2. **RAW_interactions.csv:** 1,132,367 user reviews and ratings (scale 0-5) spanning from 2000 to 2018.

RAW_recipes.csv Schema

Column Name	Data Type	Description
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name	string	Name of the recipe
id	integer	Unique recipe identifier
minutes	integer	Preparation and cooking time in minutes
contributor_id	integer	User identifier of the recipe contributor
submitted	date	Date the recipe was submitted
tags	list (string)	Categories and tags associated with the recipe
nutrition	list (float)	7-element vector: [calories (kcal), total fat (PDV), sugar (PDV), sodium (PDV), protein (PDV), saturated fat (PDV), carbohydrates (PDV)]
n_steps	integer	Number of steps in the preparation
steps	list (string)	Detailed text steps
description	string	Text description of the recipe
ingredients	list (string)	List of ingredient names
n_ingredients	integer	Number of ingredients

RAW_interactions.csv Schema

Column Name	Data Type	Description
user_id	integer	Unique user identifier
recipe_id	integer	Unique recipe identifier
date	date	Date of the interaction
rating	integer	User rating (0-5, where 0 indicates no explicit rating)
review	string	User review text

3.2 RecipeNLG Dataset

The RecipeNLG dataset contains 1,016,801 recipes and is primarily used to augment text-based generation and Named Entity Recognition (NER) tasks. It contains recipe titles, ingredients, and steps, but lacks user interaction logs, making it suitable for content-based indexing and language model fine-tuning rather than collaborative preference modeling.

4. Exploratory Data Analysis (EDA)

A comprehensive exploratory data analysis was conducted on the Food.com corpus to characterize distributions, identify outliers, and evaluate sparsity. Below are the key findings and visualizations.

4.1 Cooking Time, Steps, and Ingredients Distributions

Extreme outlier values were observed in the raw cooking time data, with some recipes reporting preparation times exceeding millions of minutes. A cooking time threshold of 1,440 minutes (24 hours) was applied to remove these outliers, retaining 229,637 recipes (99.14% of the corpus). The median number of ingredients per recipe is 9 (range 1-43), and the median number of steps is 9 (range 0-145).

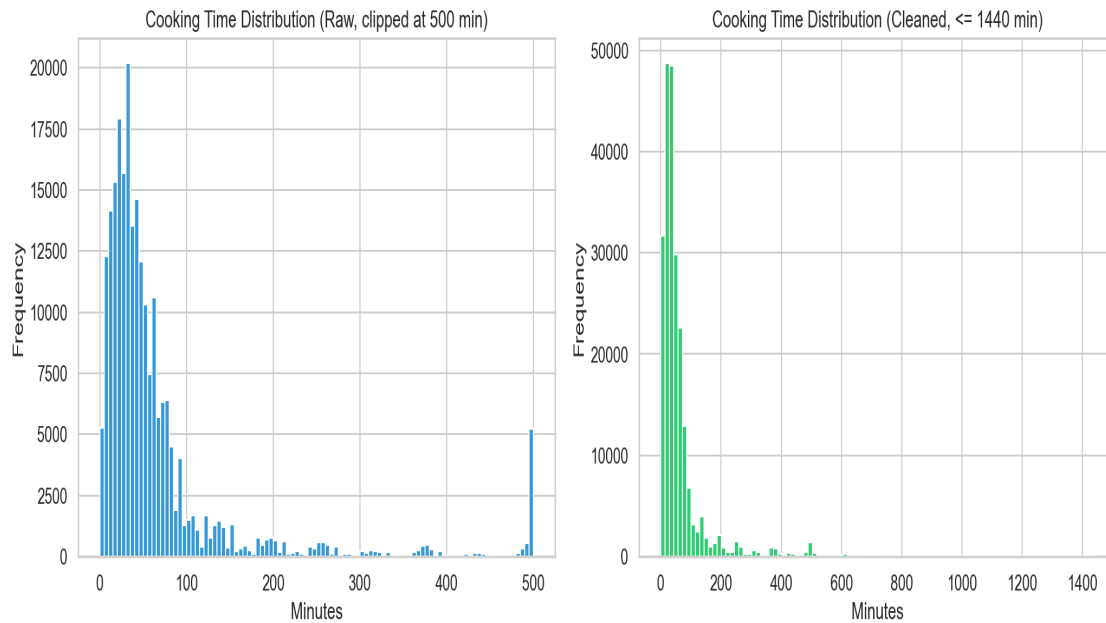


Figure 1: Cooking Time Distribution (Raw vs. Cleaned)

Figure 1: Comparison of cooking time distributions before and after outlier removal.

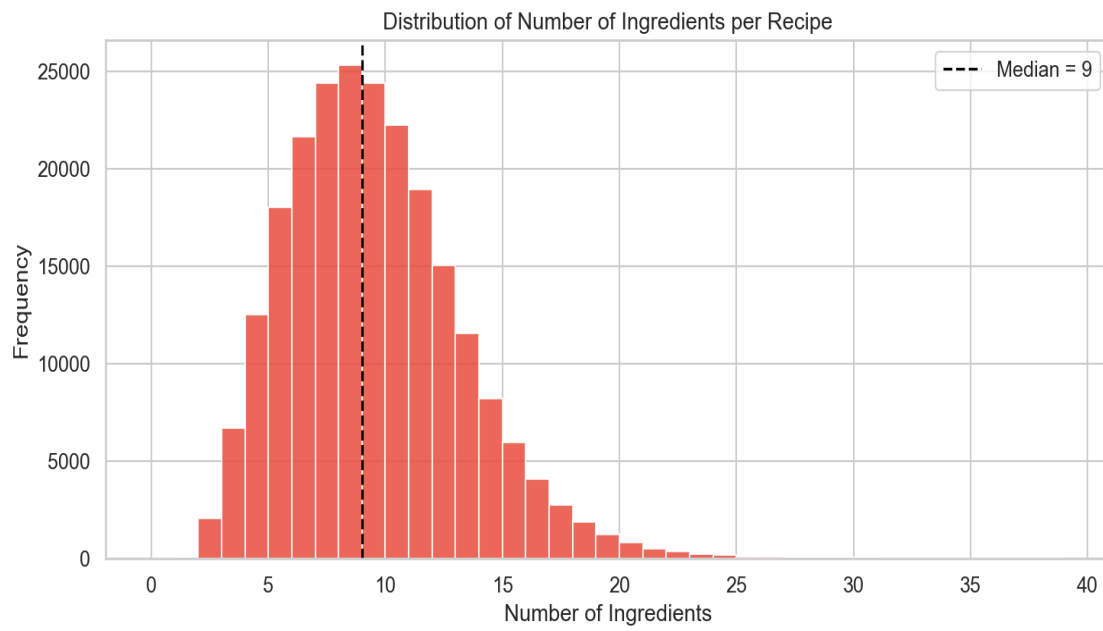


Figure 2: Ingredients Count Distribution

Figure 2: Distribution of the number of ingredients per recipe.

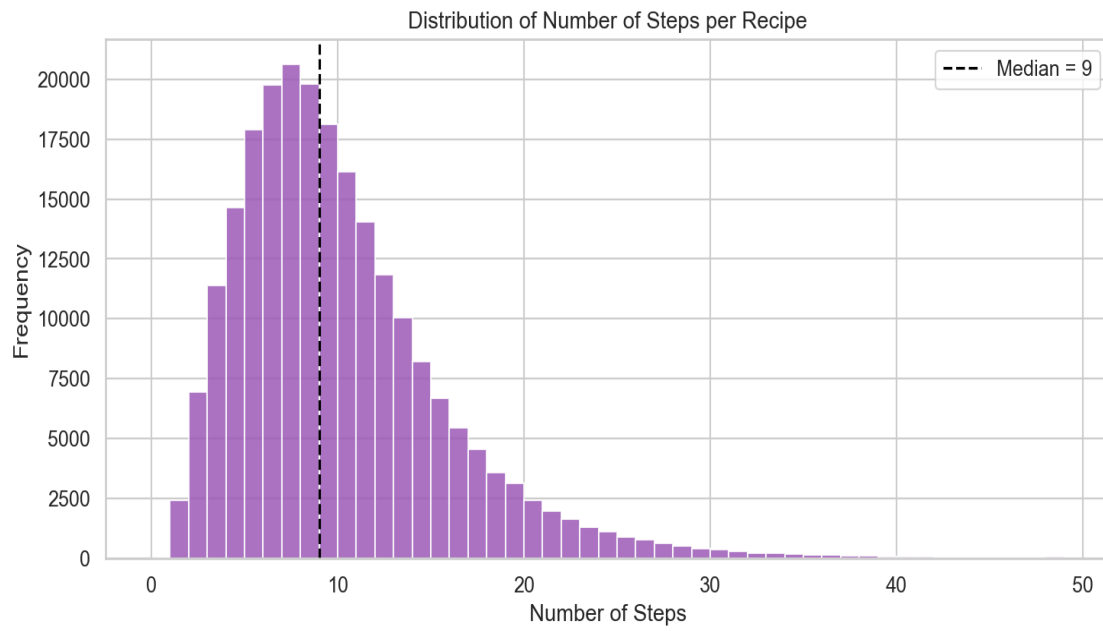


Figure 3: Steps Count Distribution

Figure 3: Distribution of the number of steps per recipe.

4.2 Nutrition Vector Analysis

The 7-element nutrition vector represents percent Daily Value (% PDV) based on a 2000 kcal diet (except calories, which is in kcal). Boxplots of these values show significant right-skew, indicating a substantial number of high-fat, high-sugar, or high-sodium recipes. Correlation analysis indicates a strong positive correlation between total fat and saturated fat (0.83) and a moderate correlation between total fat and calories (0.69).

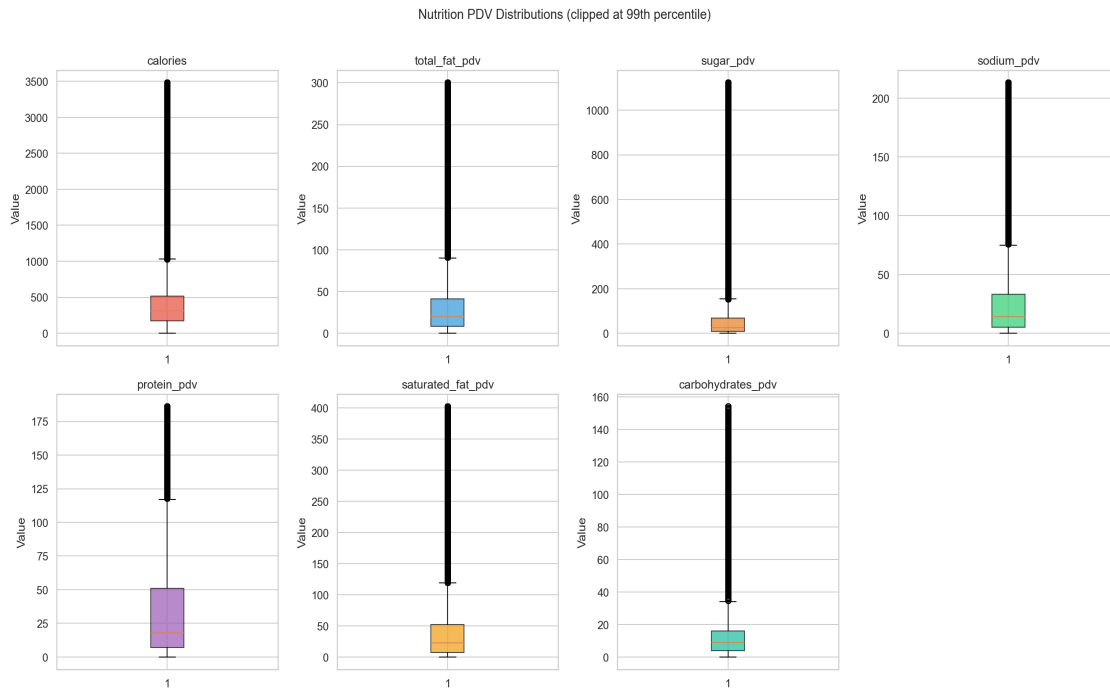


Figure 4: Nutrition Boxplots

Figure 4: Nutrition PDV distributions (clipped at the 99th percentile for visibility).

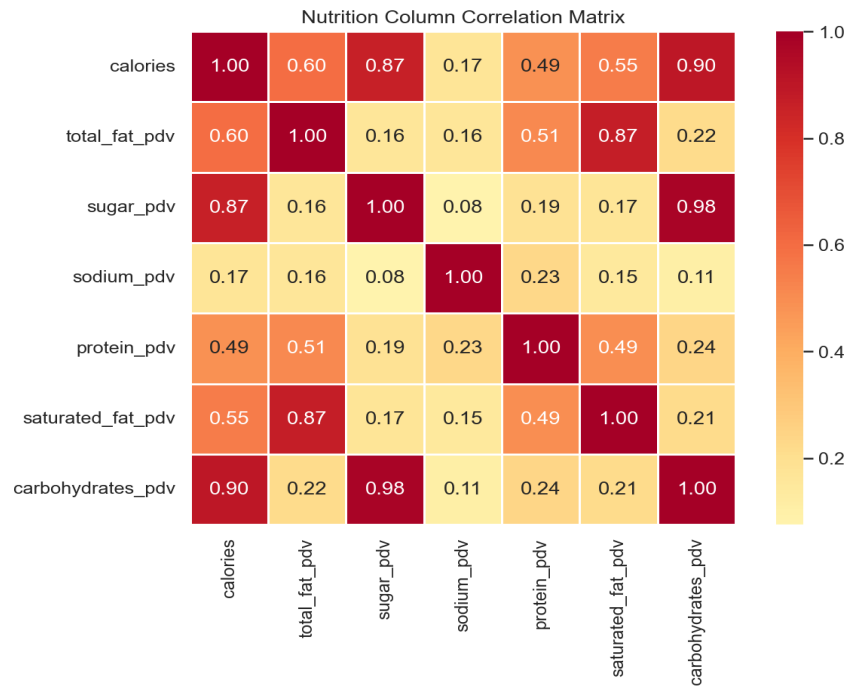


Figure 5: Nutrition Correlation Matrix

Figure 5: Pearson correlation coefficients between the 7 nutrition columns.

4.3 Interaction Skew, Activity, and Matrix Sparsity

The rating distribution is highly skewed towards positive ratings, with 5-star ratings accounting for over 70% of all interactions. Ratings of 0 represent reviews without explicit scores, which are treated as missing in explicit rating models or mapped to null in implicit feedback models.

The interaction matrix is extremely sparse:

Unique Users: 226,570

Unique Recipes: 231,637

Total Interactions: 1,132,367

Matrix Sparsity: **99.99784%**

This extreme sparsity represents a severe challenge for collaborative filtering, necessitating k-core filtering or matrix factorization to stabilize recommender baselines. Furthermore, user activity and recipe popularity follow power-law distributions, with 166,256 users (73.38%) having only 1 interaction in the history.

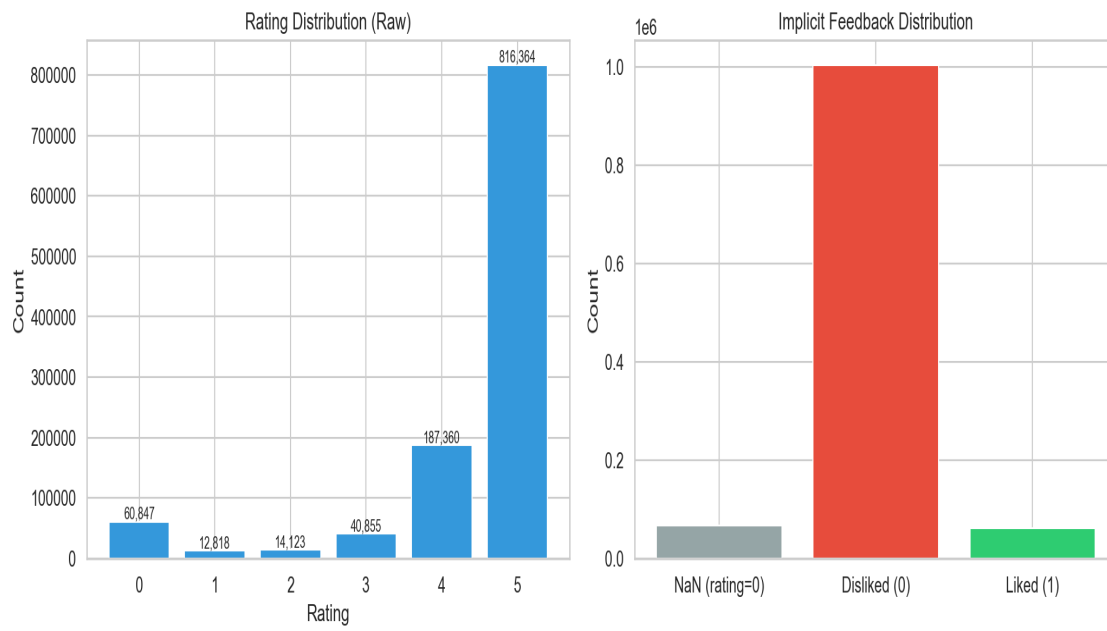


Figure 6: Rating Distribution and Implicit Mapping

Figure 6: Raw rating distribution (left) and implicit feedback mapping (right).

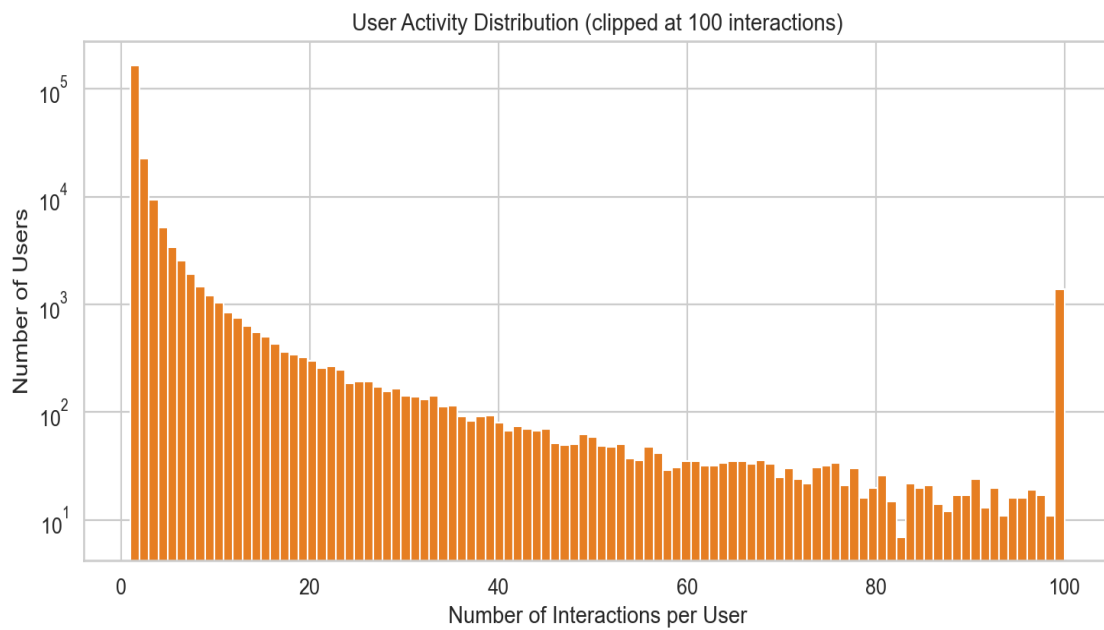


Figure 7: User Activity Distribution

Figure 7: Distribution of interactions per user on a logarithmic scale.

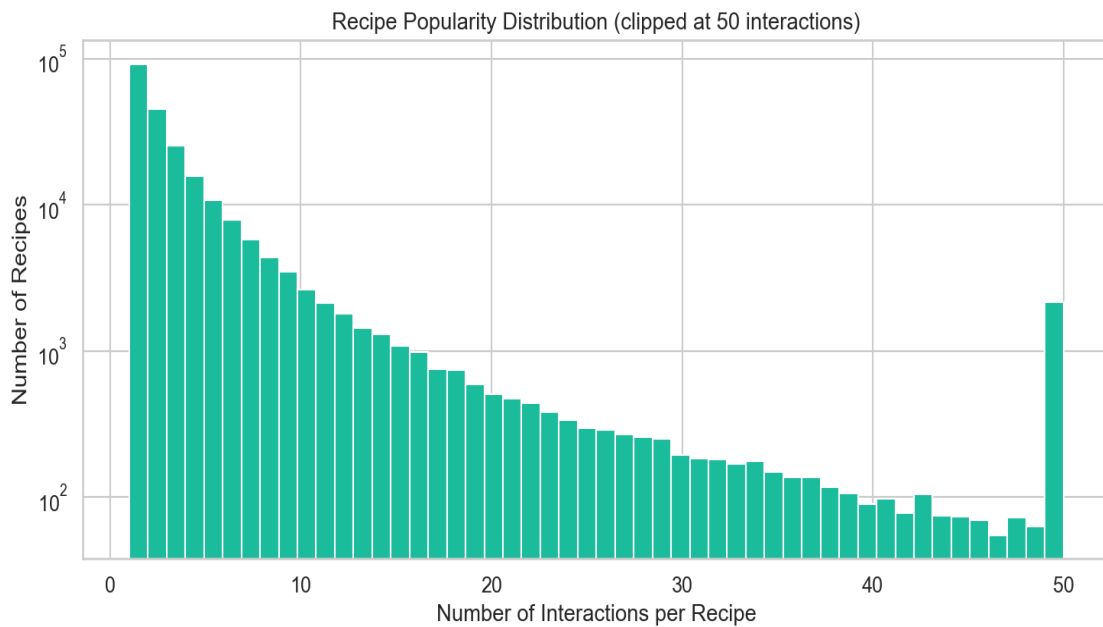


Figure 8: Recipe Popularity Distribution

Figure 8: Distribution of interactions per recipe on a logarithmic scale.

4.4 Temporal and Vocabulary Patterns

User interactions and recipe submissions span from 2000 to 2018, with submissions peaking in 2008 and user interactions peaking in 2008-2012. Tag frequency analysis reveals that "preparation", "time-to-make", and "course" are the most frequent recipe tags. The most common ingredients in the corpus are salt (appearing in 85,090 recipes), butter (54,741), and sugar (43,974).

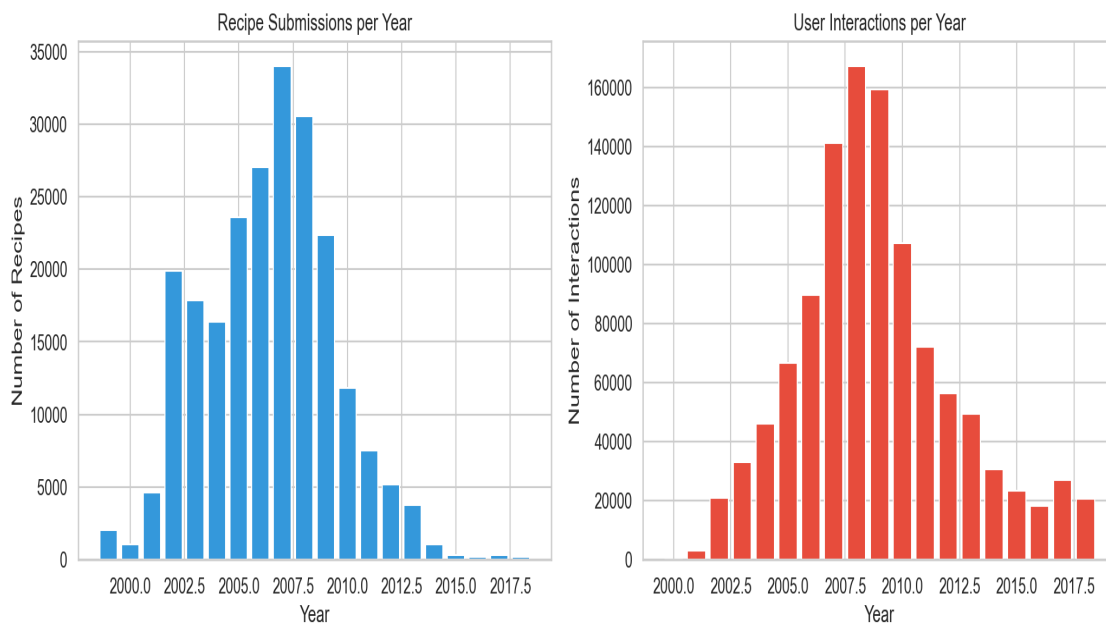


Figure 9: Temporal Patterns

Figure 9: Recipe submissions (left) and user interactions (right) per year.

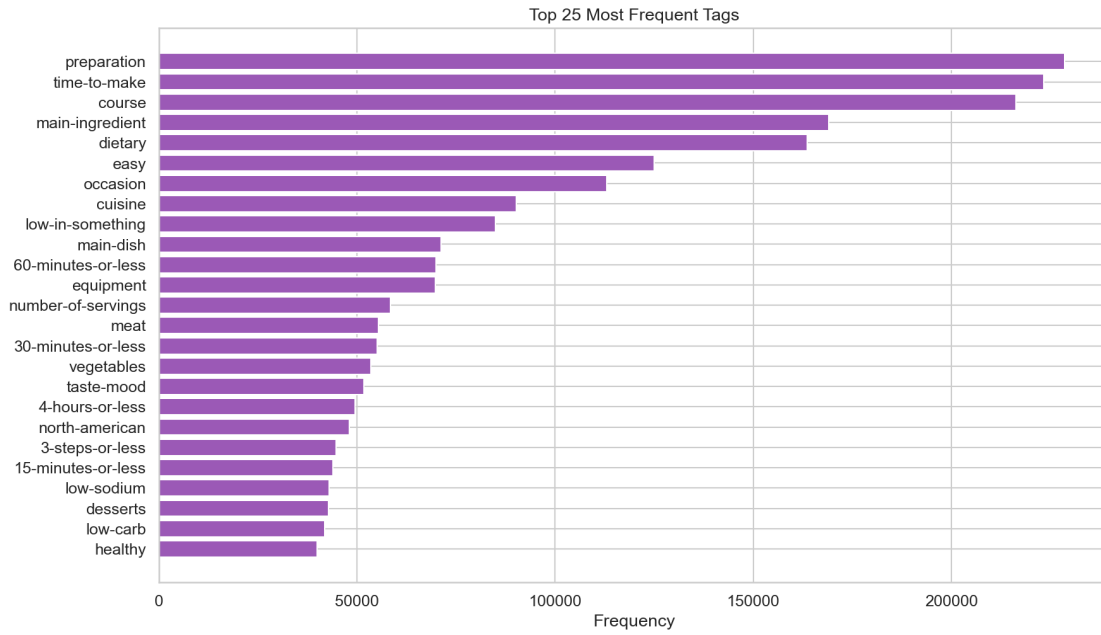


Figure 10: Top 25 Tags

Figure 10: Most frequent recipe tags in the Food.com dataset.

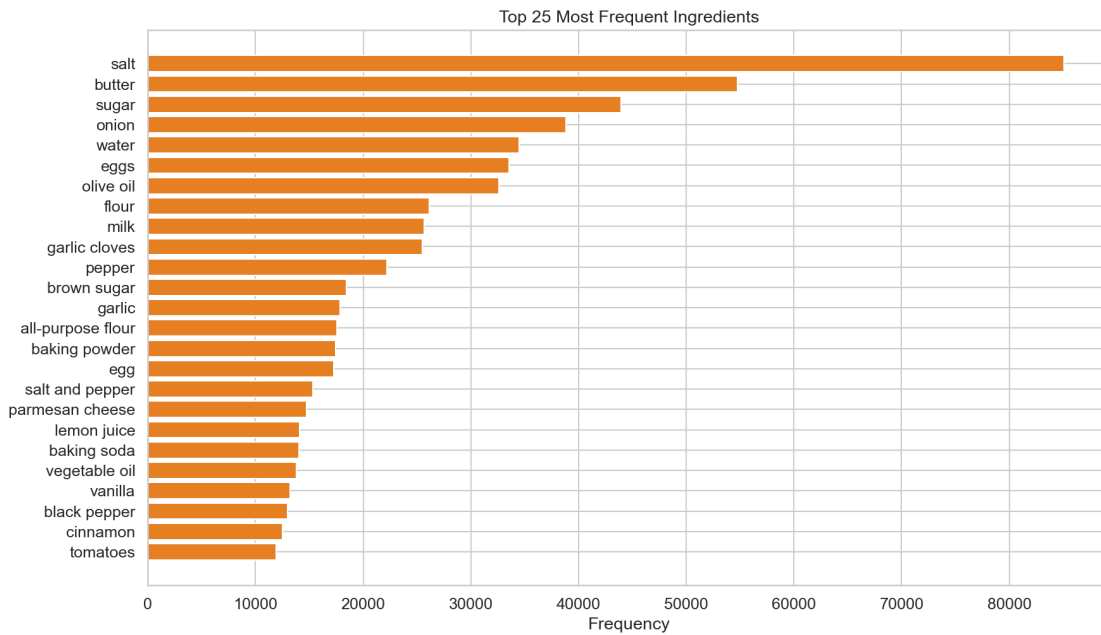


Figure 11: Top 25 Ingredients

Figure 11: Most frequent ingredients in the Food.com dataset.

5. Description of the Food.com Dataset

Following formal description guidelines, we characterize the dataset below:

5.1 Brief Introduction

The Food.com dataset is a public repository of recipe metadata and user reviews collected from the Food.com website, widely used in research for personalized food recommendation, sentiment analysis, and text-based recipe generation.

5.2 Detailed Explanation

The dataset contains structured recipes containing cooking steps, ingredient lists, and nutrient breakdowns, alongside user-wise review texts, submission timestamps, and ratings. The data spans a 18-year period, capturing evolving culinary trends and user interaction habits.

5.3 Examples

A typical entry in `RAW_recipes.csv` is "chicken lickin good pork chops" (ID 63986), which lists ingredients like pork chops, cream of chicken soup, and brown sugar, with 20 minutes of cooking time, 5 steps, and a nutritional breakdown of 289 kcal, 22% PDV fat, 14% PDV sugar, and 23% PDV protein.

5.4 Advantages

Scale: Offers over 1.1 million interactions and 230,000 recipes, making it one of the largest public food recommendation datasets.

Rich Context: Includes text reviews, nutrition breakdowns, and preparation steps, enabling content-based, collaborative, and hybrid recommender designs.

5.5 Disadvantages

Extreme Sparsity: Matrix sparsity exceeds 99.99%, causing difficulty for collaborative filtering models.

Outliers: Contains erroneous data entries (e.g., millions of minutes cooking time) that require rigorous cleaning.

5.6 Use Cases

Collaborative filtering for recipe recommendation.

Health-aware recommendation via multi-objective ranking.

Retrieval-augmented explanation generation using recipe metadata.

5.7 Limitations

Cannot be used directly for real-time recommendation without significant k-core filtering to stabilize sparse profiles.

The dataset represents a historical snapshot (2000-2018) and does not capture post-2018 culinary trends or current user dietary shifts.